SchubKI

AI Toddlers

Mind Reading Using Machine Learning and EEG

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Task and data

Which of 102 videos was this person watching, from 3 seconds of EEG?

TASK

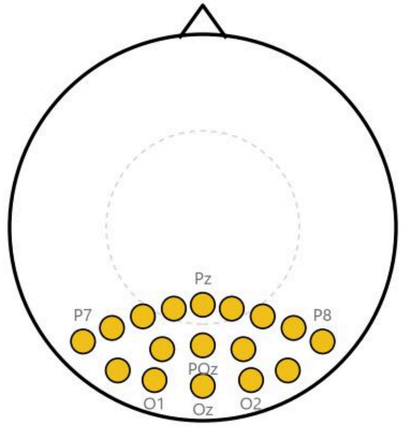
For each test recording, rank the 102 candidate videos from most to least likely.

Scored by top-1 and top-5 accuracy · chance 1.0 % and 4.9 %

DATA

6 subjects	4 development (test labels given), 2 held out (scored by the organizers)
Training	850–972 recordings per subject, one 3 s recording per video, 1,000 videos
Test	102 recordings per subject: the same 102 videos, one each, shuffled
EEG	17 posterior electrodes, 500 Hz; pre-cleaned (0.1–40 Hz, blink removal)

front of the head:
not recorded



17 electrodes over the back of the head, above visual cortex

WHAT THE DATA TOLD US

Clean, and the high frequencies are signal

no bad channels or line noise; 30–50 Hz is locked to the video

→ minimal preprocessing: 40 Hz low-pass only, no average reference

Test recordings ≈ averages of 2 repetitions

same-video correlation between subjects 2.17× higher than in training; amplitude 0.6–0.7×

→ scale each split separately; less ridge regularisation

Every test video appears once per subject

102 recordings, 102 candidates, one-to-one

→ Sinkhorn re-ranking

The information is the onset response

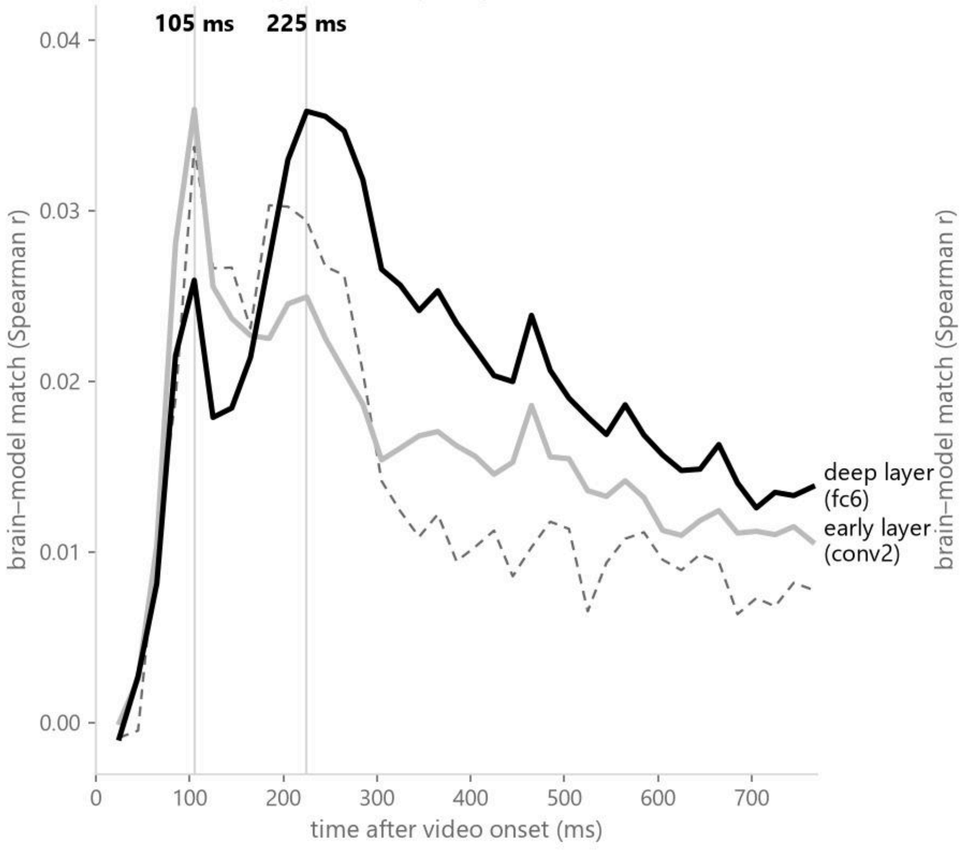
decodable 50–600 ms after onset; later EEG near chance

→ decode 50–600 ms

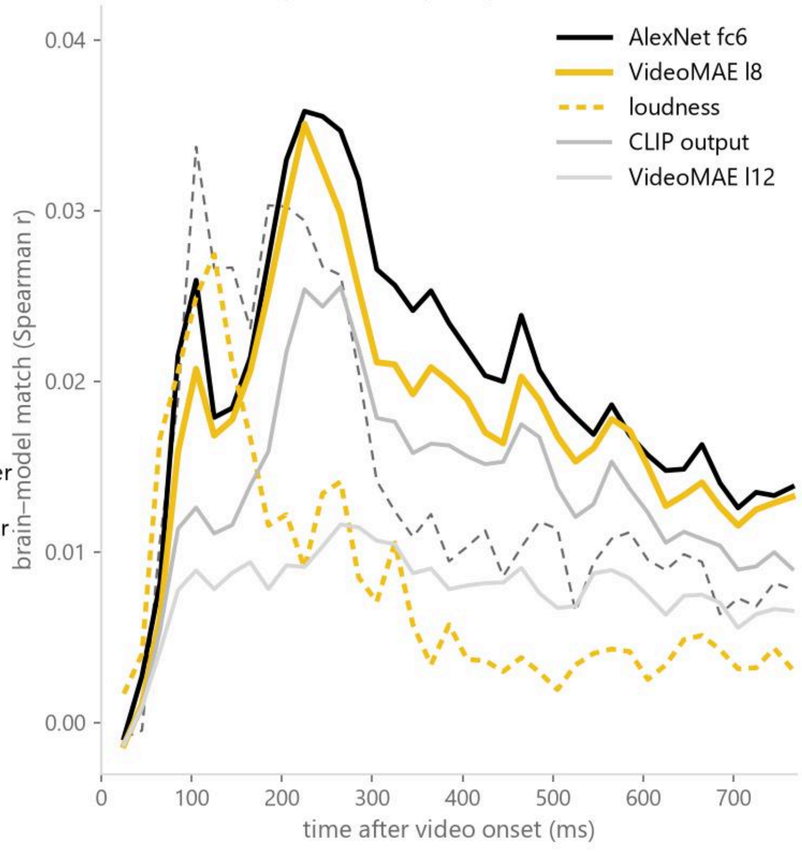
Brain vs AI: representational similarity analysis

Do the brain and a model group the same videos alike? For every pair of 695 training videos, compare how differently the EEG and the model respond, every 20 ms

1 The brain processes the video in order



2 Which models are most brain-like



1 Same order as the brain

image-network layers: early ones match at ~105 ms over the occipital pole, deep ones at ~225 ms over lateral sites

2 The EEG follows the sound

loudness matches at 125 ms as strongly as the best visual features, and independently of them

3 Mid-level beats the top

the topmost layers, trained for captions or action labels (CLIP output, VideoMAE l12), are the least brain-like

4 Brain-like targets decode better

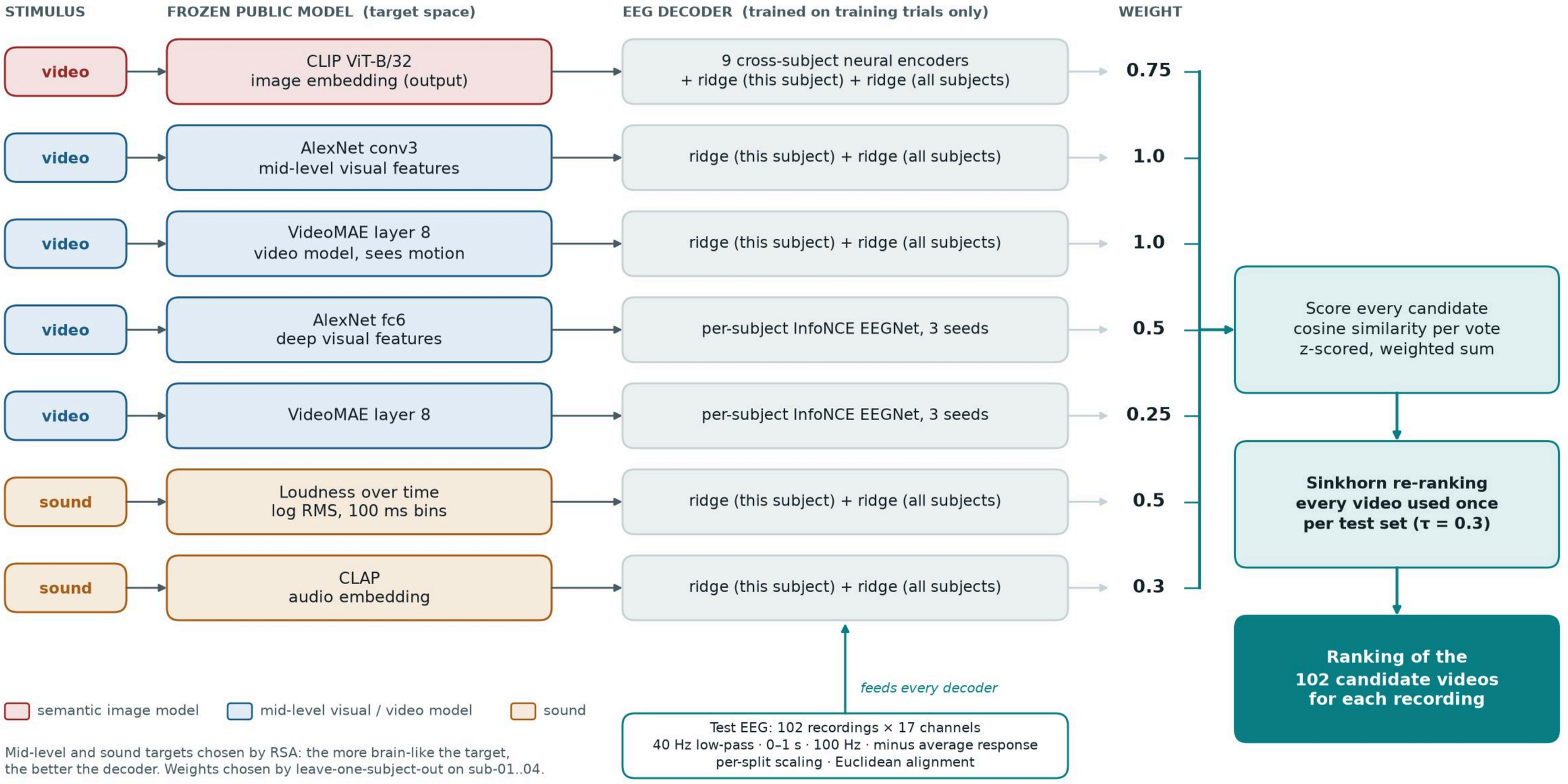
same decoder, 9 targets: the more brain-like the target, the higher the accuracy (rho = 0.69–0.80)

→ Decoding targets chosen from this:

AlexNet conv3 and fc6, VideoMAE layer 8 and loudness, the most brain-like spaces;
CLIP and CLAP kept as additional votes. Tested and rejected: V-JEPA 2 (world model) and CLIP's inner layers.

Final model: decoding EEG into the spaces of frozen video and audio models

7 votes, each a similarity between the decoded EEG and every candidate video; summed, then re-ranked jointly



Mid-level and sound targets chosen by RSA: the more brain-like the target, the better the decoder. Weights chosen by leave-one-subject-out on sub-01..04.

From evidence to design

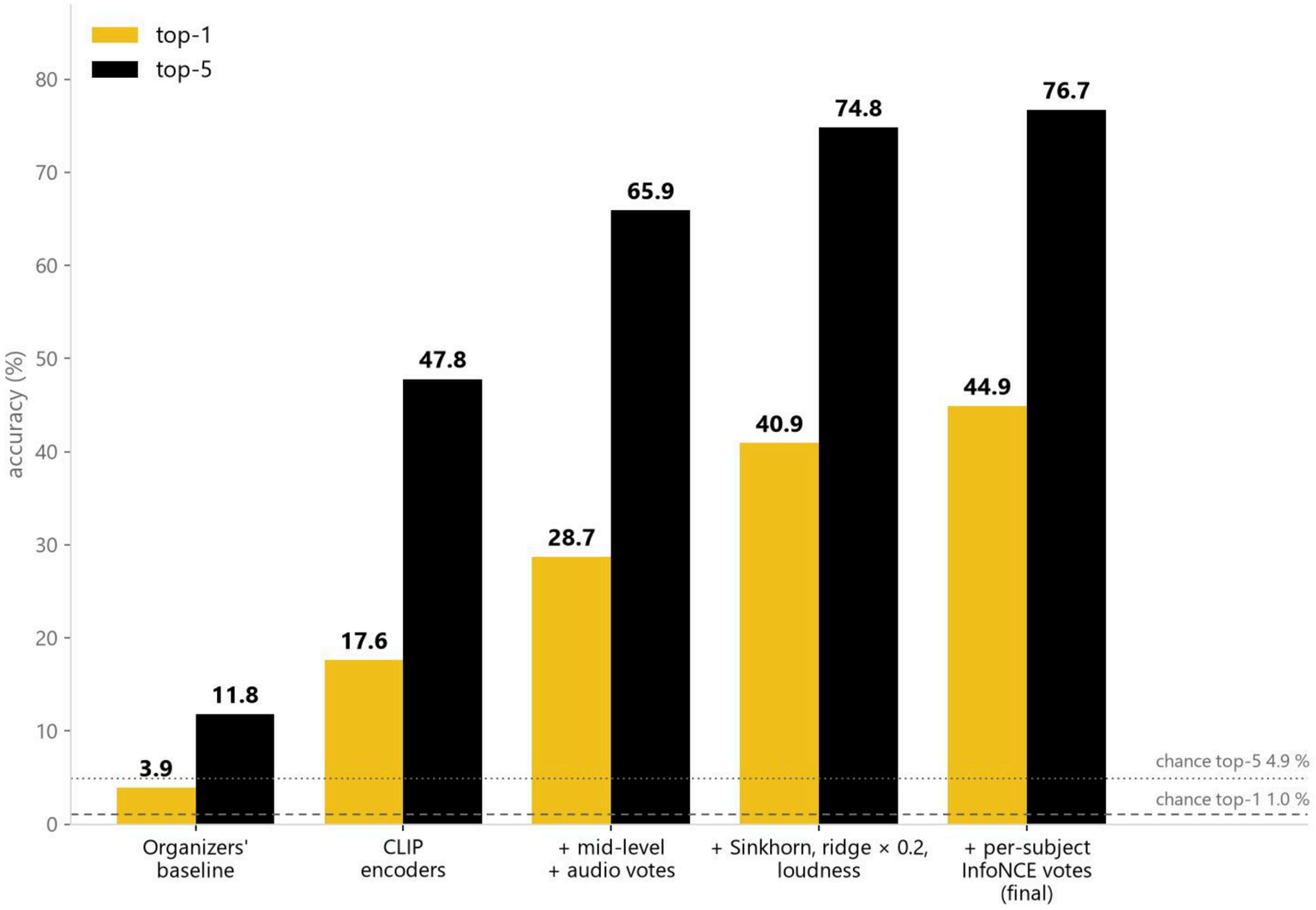
Every component of the final model traces back to a finding in the data or the brain-vs-AI analysis

SOURCE	FINDING	DESIGN CHOICE	TOP-1 LOST WITHOUT IT (dev, points)
Data checks	High frequencies carry stimulus signal; filters and average reference hurt	**Minimal preprocessing: 40 Hz low-pass only**	*basis of the pipeline (not ablated)*
Data checks	Test 0.6–0.7× training amplitude; average response removed upstream	**Scale each split separately, remove the average response**	*basis of the pipeline (not ablated)*
Data checks	Information sits in the first 600 ms (the response to the video's onset)	**Decode 50–600 ms after onset**	*basis of the pipeline (not ablated)*
Test set	Test recordings ≈ averages of 2 repetitions (2.17×)	**Ridge regularisation × 0.2**	**−4.2**
Test set	Every candidate appears once per test set	**Sinkhorn re-ranking**	**−16.0**
RSA	Mid-level layers are most brain-like; brain-likeness predicts decoding	**AlexNet conv3 / fc6 and VideoMAE layer 8 votes**	**−5.9**
RSA	The EEG follows loudness at 125 ms, independent of vision	**Loudness vote**	**−2.3**

Final model: dev 44.9 % top-1 / 76.7 % top-5 (leave-one-subject-out, chance 1.0 / 4.9 %). Bars: top-1 drop when the part is removed (mid-level: AlexNet conv3 vote; others: −2.0 to −3.0 each). Ridge × 0.2 measured on the previous version (40.9 %).

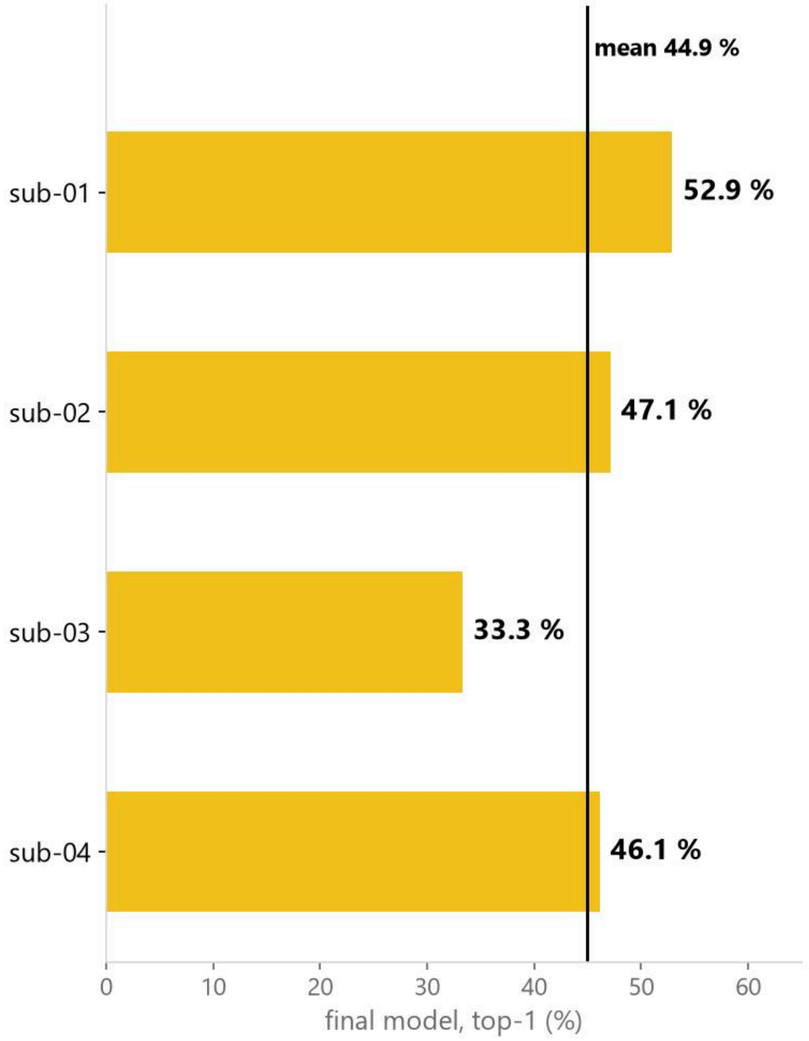
Results on the development subjects

sub-01..04: 408 test recordings, each ranked against 102 candidate videos; final model's settings all chosen by leave-one-subject-out



Baseline measured on sub-01 by the organizers. The 40.9 % version was scored with weights fitted on the dev subjects (strict leave-one-out 40.2 %).

Final model per subject



Final: 44.9 % top-1 · 76.7 % top-5 · mean rank 4.36 / 102

≈ 45× chance on top-1, 16× on top-5

What did not work, and why

Each idea was tested with leave-one-subject-out and paired tests; none gave a significant gain

TARGET MODELS

✗ **V-JEPA 2 world model**

less brain-like than AlexNet and VideoMAE in the RSA

✗ **CLIP inner layers**

better than CLIP's output, not better than AlexNet fc6

✗ **Video-to-text captions**

text is further from what posterior EEG carries

EEG SIGNAL

✗ **EEG after 600 ms**

near chance: the information is the onset response

✗ **Early EEG → early layers**

every layer carries information in both waves

✗ **Band power, covariance; average reference, whitening**

the signal is the phase-locked response; filters remove it

FUSION AND ADAPTATION

✗ **More members, transfers, score instead of rank fusion**

saturated: they reach the same EEG information

✗ **Loudness envelope alone**

too coarse to pick 1 of 102; works only as a vote

✗ **Cross-subject pseudo-averages, test-time augmentation**

no gain in leave-one-out (train and test differ)

Lesson: gains came from choosing brain-like targets and from the structure of the test set, not from adding more models or more of the signal.

How we validated, and where it stops

The numbers come from subjects the settings were not chosen on; the limits come from the data and the benchmark

VALIDATION PROTOCOL

Leave-one-subject-out on the 4 development subjects



1. **Training data only**
   models learn from the training recordings of all 6 subjects; test recordings are never trained on (challenge rules)

2. **Test labels for choices only**
   sub-01..04 test labels pick weights and settings, always on 3 subjects, scored on the 4th

3. **Paired tests**
   a change is kept only if it wins on the same 408 recordings (exact McNemar for hits, Wilcoxon for the true video's rank)

4. **A check without test data**
   RSA-chosen targets beat CLIP on 102 hidden training videos in all 6 subjects, including the held-out ones

5. **Reproducible**
   one command rebuilds the submission from the committed weights and features; all public models named with exact versions

LIMITATIONS

17 posterior electrodes
no frontal, central or temporal sites: higher-level and auditory responses are only partly visible

Noisy single training trials
one recording per training video and subject; test recordings are averages of ~2 repetitions, so train and test differ

Small evaluation
408 dev and 204 held-out recordings: differences of 2–3 points are noise; sub-06 has the fewest training trials (850)

Dev scores are optimistic
settings were chosen on the dev subjects; held-out subjects are expected to score lower

Benchmark-specific step
Sinkhorn uses that each candidate appears once per test set; an open-ended mind-reading setting would not have this

What is decoded
the visual and sound 'fingerprint' of the first 600 ms after onset, not the meaning or the story of the video

Thank you for your attention!

GitHub repository: https://github.com/LivienMajer/EEG-Mindreading